

Indian Statistical Institute

Course Name: M. Math, 2nd year

Subject Name : Topology III

Exam name : HW 4

Due: October 30, 2025

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- You may use any results proved in class. Any other results (including those in homework problem sets) require proof. You are required to provide proper mathematical reasons for your answers.

1. Let $L^\infty(C_m) \simeq BC_m$ be the infinite Lens space. Compute all the Steenrod operations on $H^*(L^\infty(C_m); \mathbb{Z}/2)$.
2. Let X be a space and $\alpha \in H^5(X; \mathbb{Z}/2)$ such that $Sq^3(\alpha) \neq 0$. Prove that $H^7(X) \neq 0$.
3. Let $\nu : S^7 \rightarrow S^4$ be the attaching map of $\mathbb{H}P^2$. Prove that ν is not a suspension, that is, there does not exist $\hat{\nu} : S^6 \rightarrow S^3$ such that $\nu \simeq \Sigma \hat{\nu}$.